

SUSHANT LOKHANDE

San Jose, CA | +1 669-340-6655 | lokhandesushant014@gmail.com | sushantlokhande.me |
linkedin.com/in/sushantlokhande14 | github.com/sushantlokhande14

EXPERIENCE

San Jose State University | Graduate Research Assistant, Applied ML & Evaluation Aug 2025 to May 2026

- Boosted malware classification to **80.2%** across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at $p < 0.001$.
- Proved a **2.4 pt** ensemble gain was real with **McNemar** testing at $p < 0.001$ instead of reporting a single accuracy number, across **1,700** held-out samples.
- Built an automated evaluation harness for controlled experiments, cutting setup effort **55%** and making every result reproducible from a single config.
- Mentored **200+ students** across **3** courses (Programming Paradigms, AI, ML) through debugging sessions, code reviews, and grading.

Turito | Software Engineering Intern, Production Reliability Jun 2022 to Oct 2022

- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Slashed p95 **CPU 22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Debugged **ICE/STUN/TURN** connection setup and degraded-network recovery across a **WebRTC** control plane and **RTP/RTCP** media paths spanning about **12 regions**.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Cut performance-investigation time about **40%** by building tooling on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.
- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL
GenAI: LLMs, RAG, grounded generation with citations, abstention, agent evaluation, OpenAI and Anthropic APIs, LangChain, PyTorch
Backend & Frontend: FastAPI, asyncio, REST, gRPC, React, Next.js
Data & Infra: PostgreSQL, Redis, Qdrant, Kafka, Temporal, Docker, AWS, pytest

PROJECTS

Document Action Agent | *React, TypeScript, FastAPI, LLMs*

- Built a document-grounded assistant that verifies sources before recommending actions and routes ambiguous cases to **human review** instead of completing them silently.

AI Learning Hub, Grounded RAG | *Python, FastAPI, LangChain, OpenAI, Docker, AWS*

- Grounded answers over **10,000+** articles with source citations and abstention when retrieved evidence was insufficient.

Teller, Agent Evaluation Environment | *Browser agents, Claude Code, Python, TypeScript*

- Analyzed **14** computer-use agent runs against an adversarial app (nested frames, missing test IDs, session expiry), classifying each failure as planning, execution, validation, or recovery.

OPEN SOURCE

- **NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)
- **microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Artificial Intelligence, Database Systems, Distributed Computing, Advanced Algorithms, NoSQL

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)